PROJECT DESIGN PHASE

Proposed Solution – Software Installation Request

ServiceNow Service Catalog • Workflow Studio / Flow Designer • Australia release

Date	21 October 2026
Team ID	
Project Name	Software Installation Request – ServiceNow
Maximum Marks	2 Marks

Project Details

Project Detail	Information
Student Name	Karra Deepak
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Project Platform	ServiceNow Service Portal / Service Catalog
Application Scope	Global
ServiceNow Release	Australia

Proposed Solution

The proposed solution uses a ServiceNow Service Catalog item and Flow Designer automation to provide employees with a consistent, traceable way to request company-approved software. The request captures the software name, version, business reason, and urgency; routes the Requested Item to the requester's manager for approval; and, when approved, creates a fulfillment Catalog Task assigned to the Software Support Team. The workflow monitors the task outcome and is designed to send the requester an appropriate status email.

Proposed Solution Template – Six Parameters

S. No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Employees need a clear, traceable way to request work-related software. Informal email, chat, or verbal requests can omit important details such as software name, version, business reason, or urgency. Approval ownership, fulfillment progress, and license availability may be unclear, creating avoidable follow-ups and delays.
2	**Idea / Solution Description**	Create the active "Software Installation Request" catalog item in the Service Catalog. Collect four answers: "What software do you need?", "Specify version (if required).", "Why do you need this software?", and "Select urgency level." In Workflow Studio, request manager approval. If approved, create a Catalog Task linked to the Requested Item and assign it to "Software Support Team." Monitor the task state and send the corresponding requester notification.
3	**Novelty / Uniqueness**	The solution connects structured catalog intake, manager approval, task creation, task-state monitoring, and requester notifications in one ServiceNow process. The "Get Catalog Variables from Software Installation Request" action has been added so submitted answers can be exposed as data pills. A separate Requested Item Business Rule, "Create Incident on License Issue," is intended to create an Incident when the Requested Item enters On Hold (state value 3).
4	**Social Impact / Customer Satisfaction**	Employees receive a standard request channel and clearer updates instead of relying on informal communication. Managers have an explicit approval step, and the support team receives an assigned task with the information needed to assess and fulfill the request. Completion and license-delay notifications are designed to set expectations and reduce avoidable status enquiries.
5	**Business Model (Revenue Model)**	This is an internal IT service-management solution, not a customer-facing revenue product. Its value is operational: improving request traceability, reducing manual coordination, and helping employees obtain approved software. No separate subscription, fee, or revenue stream is specified for this project.
6	**Scalability of the Solution**	The catalog item can be extended with more software choices, versions, request fields, and approval rules. Additional support groups, fulfillment tasks, notifications, and reports can be introduced as the service grows. The Flow Designer pattern can be reused for other IT service requests, while ServiceNow roles and access controls can support additional users and teams.

Solution Overview

The solution has four main participants. Each role contributes to a controlled request-to-fulfillment process.

User Role	Main Activities
Employee / Requester	Opens the Software Installation Request catalog item, enters the software name, version, business reason, and urgency, and receives approval and fulfillment updates.
Requester's Manager	Reviews the Requested Item approval and approves or rejects the request. The approved branch is the condition for fulfillment task creation.
Software Support Team	Receives the "Software Fulfillment Task," reviews the request details, installs the software when a license is available, and updates the Catalog Task state to reflect the outcome.
ServiceNow Administrator	Maintains the catalog item, variables, flow, assignment-group references, Business Rule, email configuration, and update set.

Catalog Item and Request Information

Question displayed to requester	Type / known internal name	Purpose
What software do you need?	Single Line Text Software_Name	Captures the software or application requested.
Specify version (if required).	Single Line Text Internal name not visible in supplied screenshot	Captures the requested software version.
Why do you need this software?	Multi Line Text Internal name not visible in supplied screenshot	Captures the business reason.
Select urgency level.	Select Box Internal name not visible in supplied screenshot	Captures the requester's urgency selection.

Solution Workflow

Stage	Activity in the proposed solution
1. Submit request	Employee submits the Software Installation Request catalog item. Four catalog variables capture software name, version, business reason, and urgency.
2. Manager approval	"Software Request Workflow" is triggered by a Service Catalog request. "Ask For Approval on Requested Item" targets Requested Item [sc_req_item] and sends approval to the requester's manager. Fulfillment proceeds only when Approval State is Approved.
3. Create fulfillment task	The approved branch creates a Catalog Task [sc_task], links it to the Requested Item, sets Short Description to "Software Fulfillment Task," and assigns it to "Software Support Team." Create Catalog Task has Wait turned OFF.
4. Retrieve catalog variables	The flow includes "Get Catalog Variables from Software Installation Request" so submitted catalog answers can be made available as data pills for later steps. The action is present in the latest workflow screenshot; its input selections and individual outputs still require verification.
5. Wait for task outcome	A separate Wait For Condition monitors the created Catalog Task until State is Closed Complete OR Pending. In this project, Pending is used as the license-unavailable signal because the available Catalog Task state choices did not include On Hold. This is a project convention, not a universal meaning of Pending.
6. Notify requester	Closed Complete follows the "Software Installation Completed" email path. Pending follows the "Software License Unavailable" path and explains that a new license must be obtained, with an expected installation time of approximately one week.

Implementation Details and Expected Outcome

Requester Notifications

Both Send Email actions were configured with Target Record = Trigger → Requested Item and Table = Requested Item [sc_req_item]. The intended recipient is the Requested for user's Email address. The requester's name is available through Requested Item Record → Requested for → Name. Catalog variables are not ordinary Requested Item columns; the flow should use the Get Catalog Variables output data pills rather than the generic "User input" field.

Notification	Subject	Intended message
Successful installation	Software Installation Completed	Personalized greeting using Requested for → Name, followed by confirmation that the requested software license has been assigned and installed. The software name is intended to come from Get Catalog Variables.
License unavailable	Software License Unavailable	Personalized greeting, explanation that the requested software license is unavailable, notice that a new license must be obtained, and an expected installation time of approximately one week. The team will proceed once a license becomes available.

Business Rule – Create Incident on License Issue

A separate Business Rule named "Create Incident on License Issue" is configured on Requested Item [sc_req_item] in the Global application. The described setup is Active = true, Advanced = true, When = after, with Insert and Update enabled. Its supplied script checks whether current.state == 3 (commented as On Hold), initializes an Incident, sets the short description to "License unavailable for " plus the Requested Item reference, adds a license-shortage description, sets Caller from the parent Request's Requested for user, sets Assignment group by display value to "Software Support," sets Priority = 3, and inserts the Incident.

Important distinction: this Business Rule checks the Requested Item [sc_req_item] state (value 3 / On Hold). The flow separately monitors the Catalog Task [sc_task] and uses Pending as its project-specific license-shortage signal. These are different records and states and must not be treated as interchangeable.

The Business Rule's Advanced-tab Condition field appeared blank in the supplied screenshot. Verify its trigger condition, reference path, group reference, and duplicate prevention with a real test before describing Incident creation as fully validated. The workflow assignment group is "Software Support Team"; the Business Rule script uses "Software Support." These displayed names are not interchangeable without confirming the actual group records.

Observed Configuration and Test Records

Item	Observed detail
Catalog Item	Software Installation Request; Application = Global; Category = Software; Catalog = Service Catalog; Active = Yes. Short description: "Request to Install company-approved software." Description: "Allows people to request software installations. Fulfills via Software Support Team."
Requested Item	RITM0010003; Request REQ0010003; Requested for System Administrator; Stage = Request Approved; State = Open.
Submitted values	Software: Microsoft Office 365; Version: 2021; Business reason: "Required for Daily work"; Urgency: High.
Catalog Task	SCTASK0010007 linked to RITM0010003; Short Description = "Software Fulfillment Task"; Assignment group = Software Support Team; State = Open.
Approval	Requester's manager is the approver. An approval record for Abraham Lincoln was shown as Approved; another row showed No Longer Required.
Update Set	"Software Installation Request Setup" in Global. The conversation first recorded In Progress with 12 customer updates; a later commit result showed Succeeded 100%, State = Committed, Collisions = 0, Total = 12.
Cross-instance check	The user reported that the full project was also successfully verified in a second ServiceNow instance (Instance B).

Expected Outcome

The proposed solution is intended to provide one traceable path from an employee's software request to the manager's decision and support-team fulfillment. It should improve the completeness of request information, make task ownership clearer, and give employees a direct notification when installation is completed or delayed by license availability. A separate Incident-creation mechanism is included for Requested Items placed On Hold due to a license issue.

Implementation status: the supplied evidence confirms the catalog item and four variables, the published flow structure, an approved Requested Item with example answers, a generated Catalog Task, and an update-set commit result. The latest workflow screenshot shows the Get Catalog Variables action is present, but its input configuration and selected output pills were not shown. The final personalized data-pill mapping and successful end-to-end delivery of both emails were not established by the screenshots. These remain verification activities, not completed claims.

Implementation Evidence

The following screenshots were supplied during the project conversation and are included as visual evidence of the configured ServiceNow components and observed records.

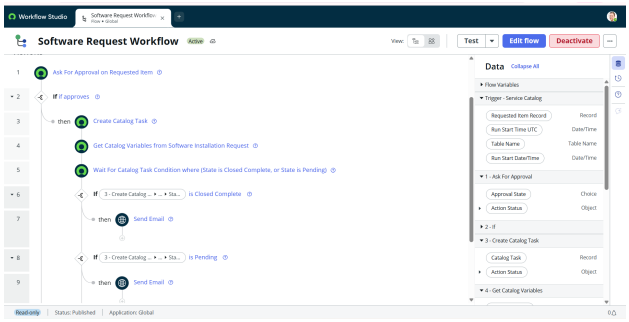


Figure 1. Published “Software Request Workflow” showing Ask For Approval, Create Catalog Task, Get Catalog Variables, Wait For Catalog Task Condition, and the two task-state email branches.

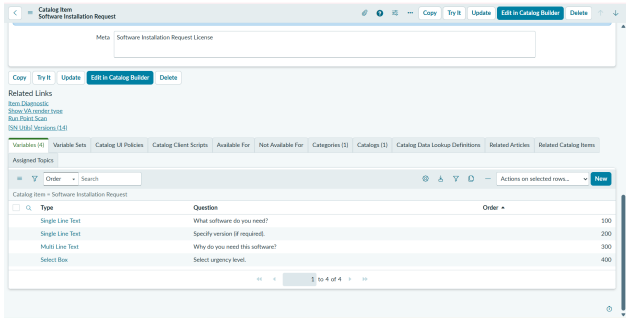


Figure 2. Software Installation Request catalog item showing the four configured catalog variables and their display questions.

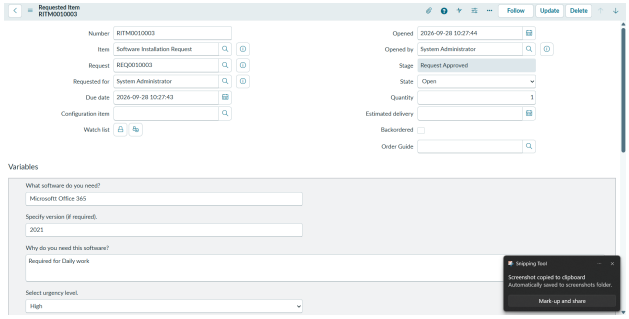


Figure 3. Requested Item RITM0010003 with example submitted values: Microsoft Office 365, version 2021, “Required for Daily work,” and urgency High.

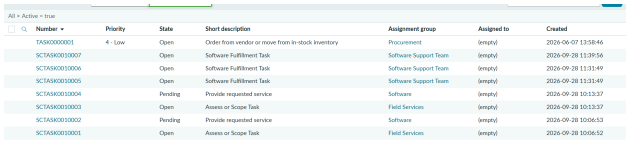


Figure 4. Catalog Task list showing Software Fulfillment Task records assigned to Software Support Team.

Evidence limitation: screenshots establish the visible configuration and record values described above. They do not, by themselves, prove that both email branches were executed successfully or that the Business Rule created exactly one correctly assigned Incident in a live test.

Source basis: project conversation and user-supplied ServiceNow screenshots; “3.2 Proposed Solution” reference PDF; and “Proposed Solution Template” PDF.